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September 15, 2015

Nature Materials

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Multi-scale metallic metamaterials

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Abstract

The smallest and largest structural length-scale of a material often differ by more than 5 orders of magnitude. Failure to span this size gap can result in materials with largely coupled mechanical properties, such as strength, ductility and density. Here we present hierarchical metallic materials with three dimensional architectures nested across seven orders of magnitude in length-scale, with structures in neighboring hierarchies separated by at least one order of magnitude. Creation of the hierarchical metamaterial with three-dimensional architectures (from centi-meter to nano-meter) is enabled by a large area additive manufacturing technique combined with a nanoscale coating process. We found that at ultralow relative densities, the hierarchical metallic metamaterials exhibit an optimal, near linear scaling between strength and density. In addition, these hierarchical metamaterials demonstrate super elasticity in both compression and tension, with high specific strength and tensile strain as large as 20%.

Main Text:

Materials with three-dimensional structures at the micro- and nanometer length-scales can often exhibit properties that are not found in their bulk material constituents, such as high stiffness and strength to mass density ratio [1, 2] [3-5], high energy densities [6, 7], high toughness, negative index of refraction [8, 9], or negative Poisson's ratio [10]. Their excellent properties can be extended into a wide a variety of applications, including energy storage and conversion [7, 11, 12], photonic devices[13, 14] and structural components, and biomedical implants. However, in efforts to make these man-made structural architectures into a macroscopic bulk form, the gap between the smallest and largest structural length-scale in these systems often spans several orders of magnitude [15-19], or are limited to a narrow window of achievable hierarchical length scales [5, 20, 21] with largest feature size extended to hundreds of micro-meters. This limit is primarily due to inadequate manufacturing and synthesis techniques. Failure to span this gap often results in a material with highly coupled properties (density, strength, stiffness, ductility and toughness) [22]. This paramount inadequacy must be addressed in order to harness micro- and nano-scale components at the macroscale.

For example, in stochastic ultralow density aerogels, the strength and stiffness degrade by more than 10^5 % times when compared to their intrinsic bulk material properties. This is due to the large pore size relative to the length-scale of graphene filaments and the low connectivity between filaments [23-26]. The recent advent of high resolution additive manufacturing has yielded mechanical metamaterials with miniaturized three-dimensional architectures [27]. These mechanical metamaterials are comprised of a periodic network of stretch-dominated architectures with stiffness to density ratio five orders of magnitude higher than that of the

aerogels. In order to reach this low density regime (typically $<10\text{mg/cm}^3$) comparable to densities of aerogels, a periodic array of thin walled nickel alloy tubes with a large diameter to nanoscale wall-thickness ratio, typically >1000 were created [15]. As a result of this large gap between controllable feature sizes at neighboring hierarchies, compressive strength of the lightest metallic lattices is reduced to less than 10^{-6} of its parent material at a relative density below 0.1%, despite their high stiffness [15, 27, 28] and strength at higher relative densities. Additionally, the tensile ductility of low density materials such as aerogels, ceramic nanolattices and metallic foams has been limited to a range of 0.3-5% [29]. The attainment of low density and high stiffness in these materials using a network of single ordered stretch-dominated microarchitectures has come with a price—significant degradation of strength and ductility at low densities.

Natural materials such as wood, nacre and bone adopt many layers of hierarchical features and length-scales to overcome this conflict and coupling of mechanical properties [30, 31]. They exhibit structures on multiple length scales of which structures at each hierarchy play a major part in determining the bulk material properties. Carefully choosing discrete length-scales and structural features at those length-scales can yield advantageous properties. For instance, trabecular bone has seven levels of hierarchy that contribute to its high strength and stiffness with minimal weight [32, 33]. It has a spongy structure; the struts or ribs in trabecular bone have a complex internal structure with microstructural hollow fibers composed of concentric lamellae. The ability of geckoes to walk up vertical walls and across ceilings is attributed to the hierarchical layout of small fibers on their toes. Each toe has a pad with hairy fibers protruding from it that repeatedly split into finer and finer hairs with dimensions down to the nanoscale

[34]. Tremendous modeling and experimental efforts have been made to mimic the structural hierarchy into man-made materials. However, these materials are still largely limited to two extremes, nanoscale lattices with maximum part size in a few hundred micro-meters, or large scale hierarchical frameworks with no structural features at the micro- and nanoscale.

In this paper, we report scalable, hierarchical metamaterials that have fractal-like, recurring architectures nested within multiple orders of magnitude comprised of length-scales from the centimeter to nanometer scales. The feature sizes in our metamaterials span more than 7 orders of magnitude, with defined three-dimensional microarchitectures separated by approximately a factor of 10 in size nested within the metamaterial. This hierarchical arrangement extends from $\sim 5\text{cm}$ to a minimal achievable feature size of $\sim 70\text{nm}$. These hierarchical materials, with stretch- and bend-dominated architectures nested in more than 5 consecutive length-scales, go beyond the limitations of all currently achievable materials at low density, and exhibit near linear, optimized scaling between strength and density. Additionally, these hierarchical metallic metamaterials exhibit ultra-high tensile ($\sim 20\%$) and compressive ductility ($>50\%$) that is not observed in any reported lightweight metal foams or lattice [29, 35-37]. Finally, at overall part sizes approaching the tens of centimeter range, these unique materials will begin to become practical for many applications.

We refer to hierarchical order n to represent the total level of scale with recognized structure within the material. For $n = 0$, the material is viewed as a continuum for the purpose of analysis of physical properties; $n = 1$ (first order) represents a latticework of continuous ribs analogous to the atomic lattice of a crystal. Likewise, $n = N$ represents a higher order latticework that is

comprised of architectures at the $(N-1)^{\text{th}}$ level. On each level of hierarchy, orientation of unit cells, relative density, tessellation, and feature sizes are designed and repeated in 3D space to comprise the fractal lattice.

An example of a fractal-like material that contains length-scales spanning five orders of magnitude with more than three orders of structural hierarchy is shown in Figure 1. The stretch-dominated unit cells which carry load through tension or compression are distributed across three hierarchical levels. Each level of hierarchy has feature sizes which differ by at least one order of magnitude. On the macroscale ($\sim 5\text{cm}$) (Fig 1A), the bulk lattice material is comprised of a network of second order ($N=2$) hierarchical stretch-dominated octahedron-tetrahedron (octet) unit cells (Fig 1B). A second order nickel-phosphorous octet unit cell (Fig 1 C and D) contains $\sim 200\mu\text{m}$ hierarchical strut members (Fig 1 E) that are comprised of a network of first order octet unit cells organized in the (110) orientation (Fig. 1F). These first order unit cells are comprised of micro-scale thin-walled struts (Fig. 1 G). At the lowest hierarchy (Fig. 1 G and H), these materials are made out of nickel-phosphorous thin-walled hollow tubes with a thickness ranging from 70nm to 1 μm . Additional fractal hierarchy can be created as shown in Fig. 1 (I)-(L); a 3rd order stretch dominated octet unit cell with struts comprised of second order octet lattices.

The connections between unit cells are considered rigid with relative orientations (referred to as Miller Indices at each hierarchical order and tessellation methods that are adjustable through a customized script that generates the fractal lattice architectures (See **supporting information**)). At each level of hierarchy, the microarchitectures can be defined and spatially arranged along a path defined by the architecture of the higher level. Relative density, unit cell classification, cell

sizes, and aspect ratios of ligaments in each hierarchy can be deterministically defined. Design variables associated with the ligaments in each level of hierarchy are, t , d_1 , l_1 , d_2 , l_2 , d_3 , l_3 , as shown in Figure S2, thereby expanding the usable design variables beyond a single length-scale. A variety of stretch-dominated fractal unit cell designs achieving the same overall relative density were generated using the fractal lattice generator, see Figure S2-S3.

Additionally, combinations of structural architectures can be created by constructing one hierarchy with a differing unit cell. For simplicity, microarchitectures at each hierarchical level are categorized as stretch dominated and bend-dominated to make up the basic palette of the fractal material generator (Basic palette is shown in Figure 2 (A) and (B)). A cubic and beam shaped hierarchical lattice can be readily constructed by tessellating two microarchitectures along two orders of hierarchical length-scales, with a bend-dominated architecture as the second order and a stretch-dominated as the first order (Figure 2 (C) and (D)). Figure 2 (E) – (G) shows the additively manufactured nickel-phosphorous bend-stretch hierarchical metamaterial, with hollow-tubes as their lowest structural hierarchy.

The fabrication of the hierarchical metamaterials is enabled by a scalable, high resolution, projection micro-stereolithography process combined with post processing techniques [38, 39]. The process begins with template fabrication of solid polymer fractal structures up to the third order of hierarchy. Two-dimensional images of fractal patterns are sequentially transmitted to a reflective spatial light modulator (SLM), which is illuminated with UV light from a light emitting diode array. Each image is projected through a reduction lens onto the surface of the photosensitive resin. The exposed liquid cures, forming a layer in the shape of the two-

dimensional image. After a layer is formed, the substrate on which it rests is lowered, fresh resin flows over the structure forming a thin film of liquid over the cured layer. This process is repeated layer-by-layer until a complete structure is formed. The additional order of structural hierarchy with features down to 50-100nm was obtained by further processing of the architected polymer. Using the base polymer lattice as a template, we coated the structures with a thin layer of nickel-phosphorous shell via electroless nickel plating. The thickness of the metal coating is controlled by the plating time, yielding metal films from 70 nm to 1 μ m. Once the nickel shell grew to a desired thickness, the polymer structure was chemically etched from the structure leaving behind hollow-tube lattices with the smallest structural features at the 70-200nm level. A detailed description of the nickel-plating procedure is given in **Materials and methods**. XRD, TEM and energy dispersive x-ray spectroscopy confirmed that the film composition is 11% phosphorous and 89% Nickel by weight, with the XRD peak corresponding to a brittle like behavior of the nickel phosphorous [40-42]. The resulting hierarchical materials contained sub-100nm features in centimeter sized components, with successive reduction of feature size by at least one order of magnitude per hierarchical level. With fractal hierarchical design, hierarchical polymers with relative density ranging from 11% to 2% were fabricated and hollow tube nickel lattices with relative density 0.1% to 0.012% were created.

In contrast to first order lattice materials whose compressive strength-density scaling becomes quadratic or even cubic at ultra-low densities [4, 27], hierarchical lattices at the same relative densities have a tunable scaling relationship and diverse failure modes. This is because the failure mode with the lowest critical load governs the overall stability of the hierarchical lattice. The critical buckling and yielding loads of structural features situated at different levels of hierarchy contribute to the overall stability of the mechanical metamaterial. This can be

characterized by the ratio of the first order nanoscale wall thickness to ligament diameter ratio and second order filament aspect ratios, t/d_1 and d_2/l_2 , respectively. Two common failure modes of hierarchical lattices are summarized in Figure 3 A-B. For second order stretch dominated lattices with ligament parameters identified in TABLE 1 of the Supplementary Material, the uniaxial compression test shown in Figure 3A demonstrates local buckling (I) of first order tubes, followed by a pseudo “plastic” deformation section, as the compression load continues after first order tube buckles, the “second order” buckling occurs at (II) as viewed from the snapping behavior of the lattice (Figure 3A and MOVIE). The snapping event as a result of elastic buckling of the second order beam is indicated by a negative stiffness section following the turning point on the load-deflection curve. On all subsequent load and unloading with maximum strain value over 60%, the same successive “first order--- second order” alternation buckling occurs, with complete elastic recovery.

For lattice structures in Figure 3B, where effective stress for second order beam buckling is lower than that of the first order hollow tube buckling, Euler buckling at the second order beam occurs. The post elastic buckling behavior (snapping of second order struts) of the second order beam corresponds to the negative stiffness region in the loading following the elastic buckling event. No damage is observed with 98% recovery after the load is applied to 60% strain.

This diverse deformation modes of hierarchical lattices is attributed to the interplay between distinct failure modes within each level of hierarchy: local buckling of the walls, Euler Buckling of the first order tubes, yielding/fracturing of the first order struts, and Euler Buckling of the second order lattice. The full derivation of the failure modes among hierarchies is detailed in the **Supplementary Materials**. A contour map between hierarchical features along with their overall relative densities and failure mode is plotted in Figure 3C. The ratio of the wall thickness to first

order diameter (t/d_1) and the aspect ratio of the second order struts (d_2/l_2) are plotted with a failure mode map in Figure 3C. The relative density of the hierarchical metamaterial is correlated with t/d_1 and t/d_1 as derived in the supplementary material. For a given density as indicated by the relative density contour lines, the maximum compressive strength occurs when the critical strength for first and second order struts approach the same value. In regions above this line, first order Shell buckling is the dominated failure mode whereas in regions below the line Euler Buckling is the dominated failure mode. This was confirmed by Finite Element Analysis using the NIKE3D software package (see **supplementary information**), where the hierarchical periodic unit cell is modeled and the two distinct buckling modes are shown in the inset of Figure 3C. Ultralow relative density lattices that fall along the optimal line where the two failure modes intersect offer the maximum specific strength at a given relative density. These fractal-like stretch-dominated metamaterials along the optimal line exhibit unprecedented scaling between yield strength and relative density with a scaling power of 1.3. This far exceeds first order octet metamaterials, which have a scaling power over 2.5 observed in Ni-P microlattices at relative density below 0.2% [43]. Consequently, the hierarchical lattice experiences yield stresses two orders of magnitude higher than that of ultralow density first order periodic lattices.

Even more remarkably, hierarchical metamaterials with combinations of different architectures exhibit high tensile and compressive ductility despite their relatively brittle constituent material. Uniaxial compression tests of the hierarchical bend-stretch dominated lattice illustrated in the Supplementary Material exhibits elastic deformation as large as 58% after multiple loading and complete unloading cycles as shown in Figure SX.

Uniaxial tensile tests on bend-stretch dominated hierarchical lattices showed unprecedented high tensile ductility. Fig. 4 A, B, C are tensile tests of bend-stretch hierarchical lattices with nanoscale wall thickness of 1 μ m, 200nm, and 70nm. When the material is stretched, the second order hierarchical ligaments rotate about their nodes and start to unfold from their original position, which contributes to the toe-region of the curve. As the load is further increased, a continuously increasing amount of first order strut members start to rotate about their nodes until the first order struts members form a near 90 angle from the horizontal position and continue to be uniaxially stretched. The collective rotation of second and first order strut members under global tensile stress can be seen from the stretching of the open pores within the second and first order lattices in Figure 4. The largest amount of second order and first order pore stretching was observed in lattices with 70nm wall thickness. As most first order strut members are stretched, a linearly increasing stress versus strain response is exhibited. At larger strains, the hierarchical lattice undergoes micro-damage in the nodes of the first order strut members, leading to propagation of node breakings (plateau section) and fracturing of the second order ligaments (Supplementary MOVIE). This damage ultimately leads to tensile failure of the material. We found that the maximum tensile strength is roughly 25-40% of the compressive plateau stress. Given the relative brittle nature of the Ni-P constituent material, no plastic or strain hardening was observed. In contrast, at a coating thickness exceeding 500 μ m, the uniaxial tensile behavior resembles that of conventional metallic foams where fracture is observed at a strain less than 7%. Figure 4D compares hierarchical lattice tensile properties, with other recently reported ultra-light materials and foams. At a relative density below 0.1%, the specific tensile strength of this nickel alloy hierarchical material reaches a value of 40.8 MPa/g-cm³ with a linear strain approaching

20% for materials with the lowest wall thickness (70nm). This specific strength relative to the density of the lattices is two orders of magnitude higher than alumina foams and lattices [44-46].

We demonstrated the creation of metallic metamaterials with hierarchical three dimensional architectures nested within 7 orders of magnitude in size. These hierarchical metamaterials, with configurable recurring structural architectures and nanoscale wall thickness at the lowest hierarchy exhibit simultaneously high strength, stiffness and ductility in the ultra-low density regime. The flexible, hierarchical architectural design is enabled by a fractal lattice generator that incorporates a library of three-dimensional microarchitecture unit cells which can be inserted across multiple orders of length-scale. The fabrication of the hierarchical metamaterials is achieved by a large area additive manufacturing technique combined with a nanoscale material metallization process. In contrast to stochastic and periodic low density materials where a large structural gap persists in the length scales of structural features, hierarchical metamaterials incorporate fractal or hybrid of architectures from the centi-meter to nano-meter length-scale. Through optimizing structural variables among multiple hierarchies, hierarchical metamaterials of the 2nd and 3rd order exhibit a scaling relationship between strength and density as low as 1.3. In addition to high specific strength, bend-stretch metallic metamaterials exhibit ultra-high elastic limits (10-20%). This is surprising because in bulk and cellular metal alloys, a low elastic limit (<5%) typically results from plastic yielding or brittle fracture. These results point to a new material design paradigm where material properties such as density, ductility and strength can be co-designed from structural architectures spanning successive length scales inspired by nature's hierarchical materials.

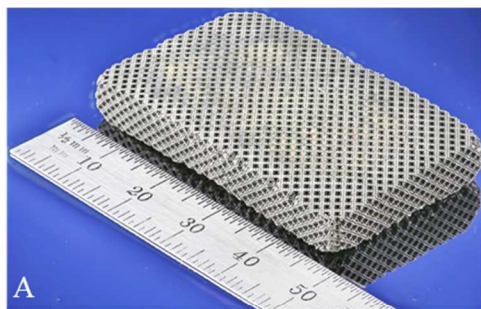
Fig. 1. Nickel alloy fractal structural lattice. (A)-(C) Optical microscope images of bulk hierarchical lattice material with a network of hierarchical stretch-dominated octet unit cells. (D)-(H) Scanning electron micrographs (SEM) showing breakdown of structural hierarchy of a second order lattice unit cell (I) Optical image of a 3rd order nickel-phosphorous, hollow tube unit cell; (J)-(L) SEM showing breakdown of structural hierarchy of a third order lattice unit cell

Fig. 2. Hierarchical stretch-bend lattices. (A) Schematic illustration of combinations of structural hierarchies from stretch and bend dominated architectures. (B)-(C) Bend-stretch dominated lattice tensile specimen and cubic bulk specimen.

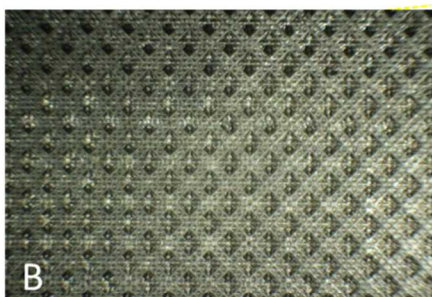
Fig. 3. Tunable compression response of fractal nickel-phosphorous hierarchical lattices. (A) Fractal lattice exhibiting first order buckling followed by second order buckling. (B) Fractal lattice compressive response exhibiting second order dominated Euler buckling without structural damage on the first order. (C) Failure map of ultralight hierarchical octet lattice as a function of first order and second order design parameters. The red curve shows optimal strength at a given relative density. Each contour line represents values of relative density. (D) Strength-relative density scaling of stretch dominated hierarchical lattices compared against non-hierarchical lattices and literature data.

Fig. 4 Tensile response of hierarchical lattices (A) Still frames and mechanical test data from the uniaxial tensile test on a thin-walled hierarchical (bend-stretch) lattice showing high tensile ductility with tensile strain approaching 20%. (B) Mechanical data and still frames from the uniaxial tensile tests on a thick-walled hierarchical (bend-stretch) lattice demonstrating low elasticity, plastic and fracture behaviors similar to conventional metal foams. (C) Mechanical data and still frames from uniaxial tensile tests on a thin-walled hierarchical (bend-stretch) lattice demonstrating transitional behavior -- higher ductility followed by brittle fracture. (D) Specific

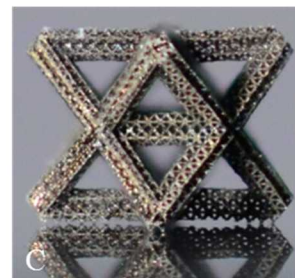
ultimate tensile strength versus elongation to fracture of hierarchical bend-stretch lattice material compared with non-hierarchical stretch-dominated nickel lattices and available literature data on ultralight material tensile properties.



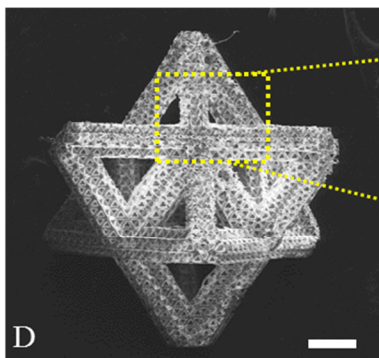
A NiP hierarchical metamaterial ~5cm



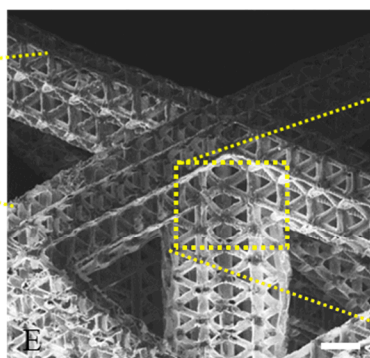
B 2nd order lattice ~5cm



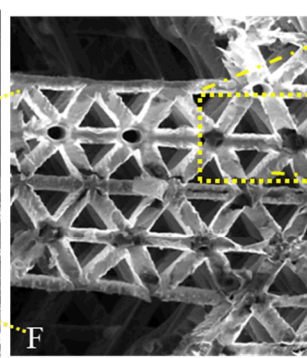
C 2nd order cell ~2mm



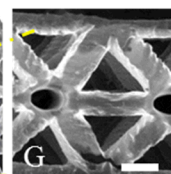
D 2nd order unit cell ~2mm



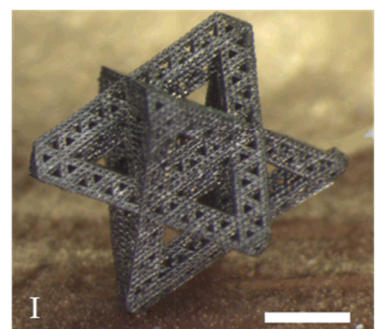
E 2nd order strut ~200μm



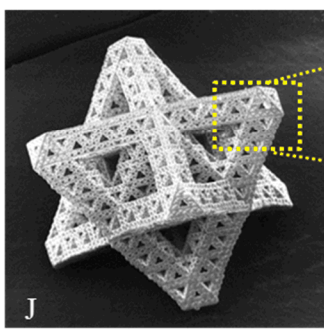
F First order lattice 20μm



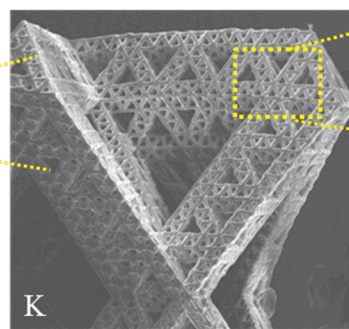
G hollow tube 80nm



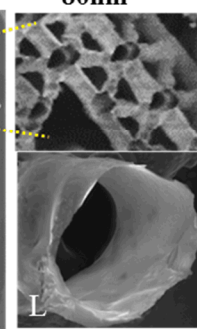
I 3rd order unit cell ~2mm



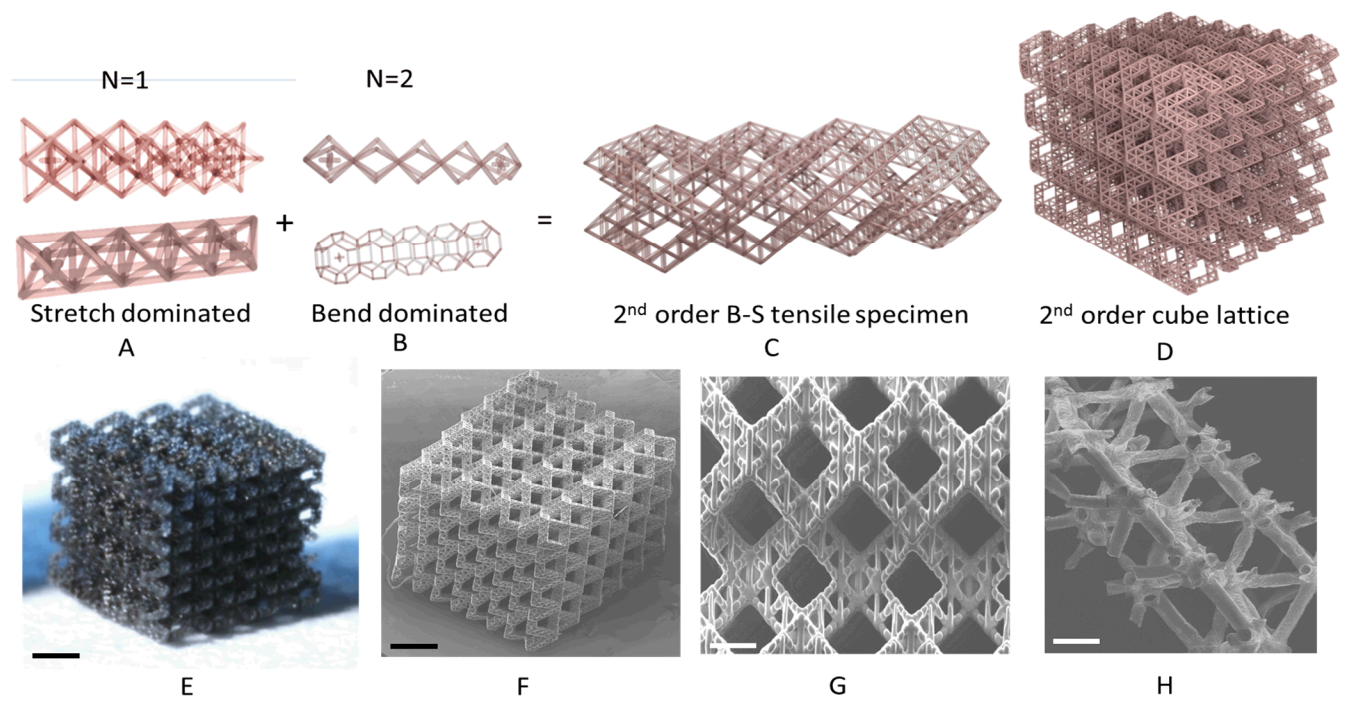
J 3rd order unit cell ~2mm

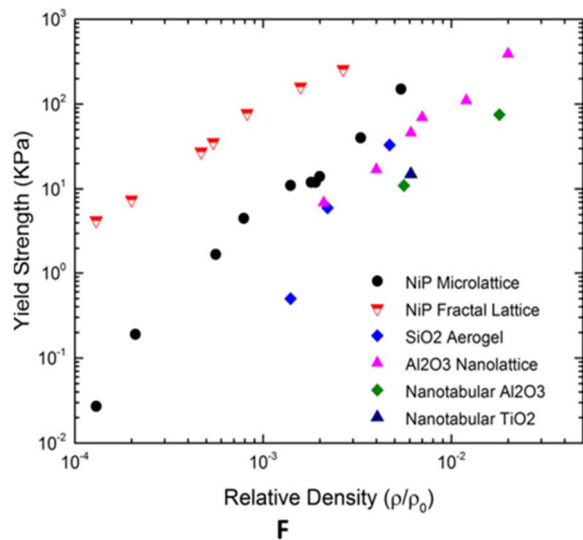
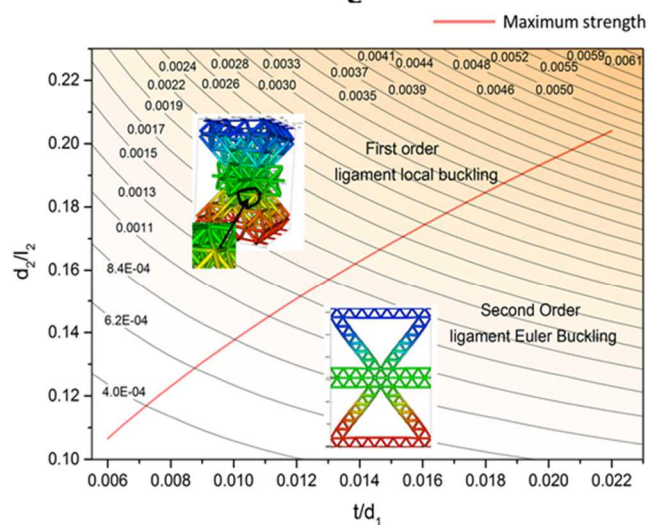
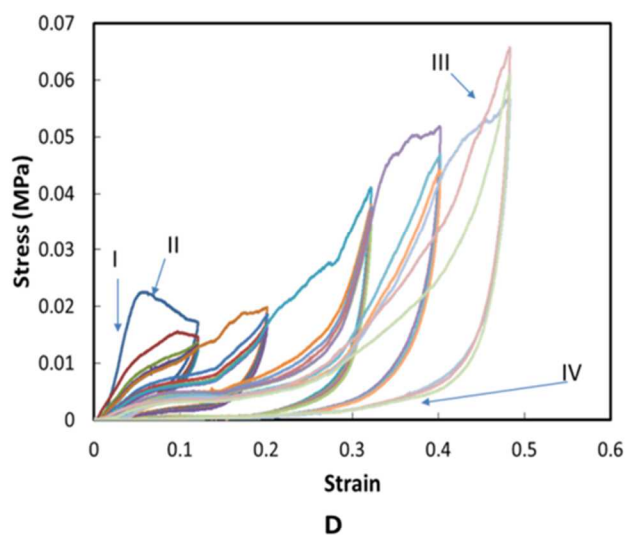
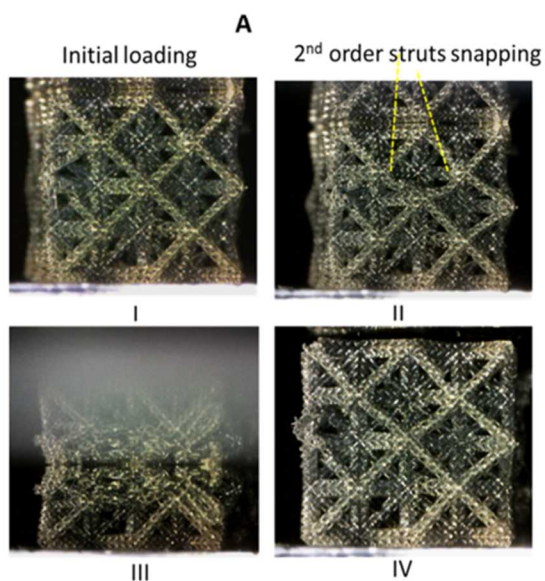
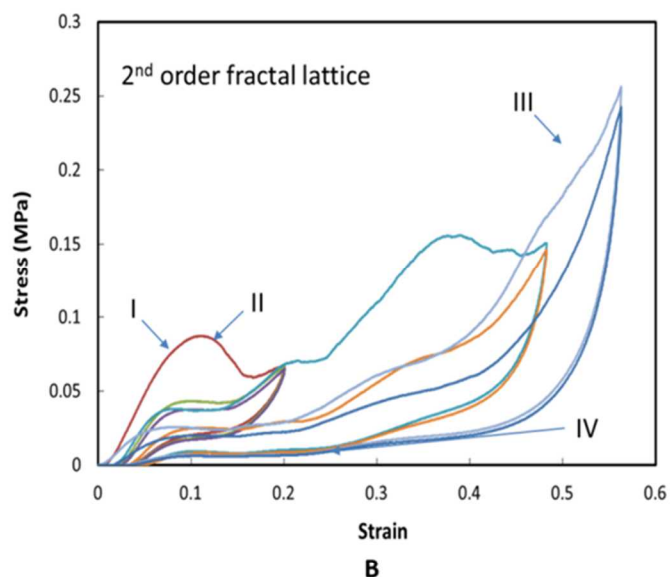
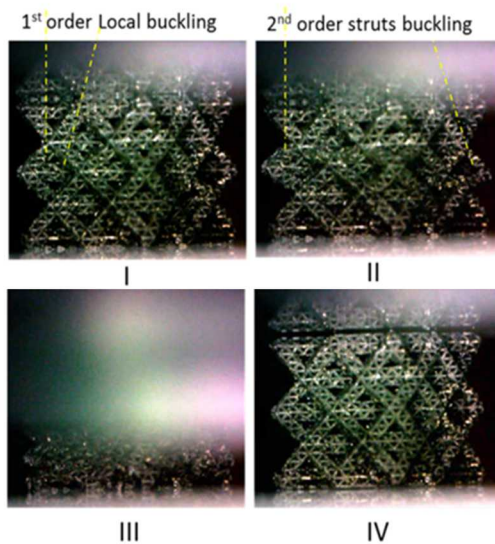


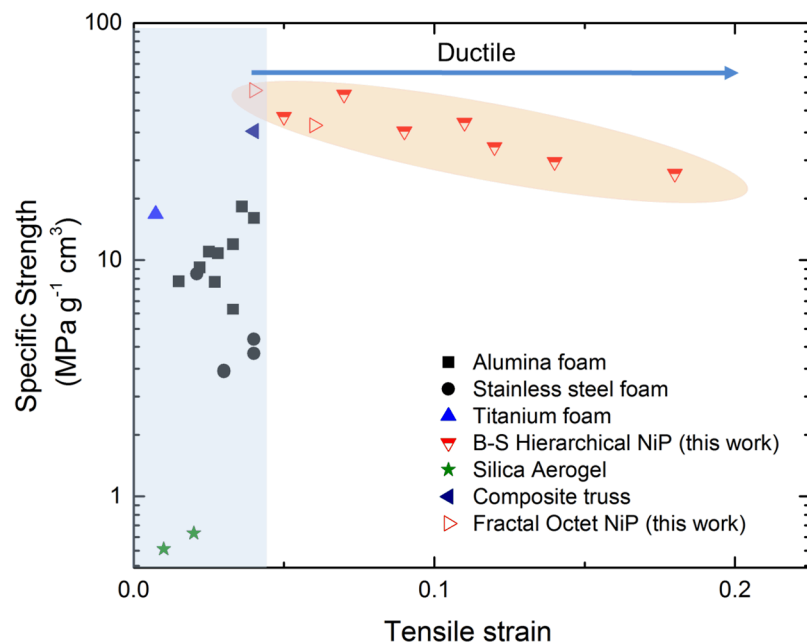
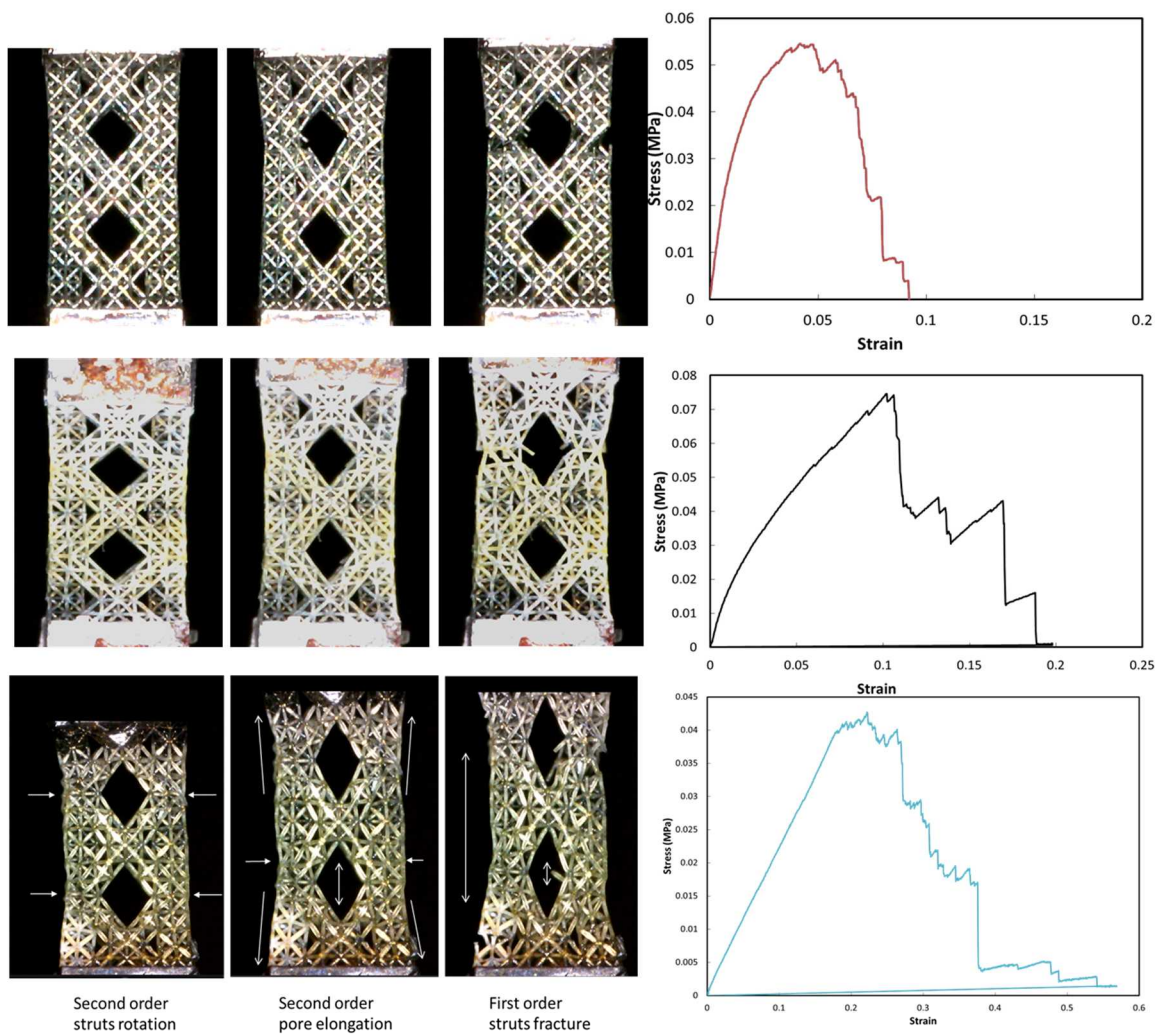
K 3rd order strut ~200μm



L Hollow tube 5μm~70nm







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SUPPLEMENTARY MATERIAL

S1 Materials and Methods

S1 .1 Additive manufacturing of hierarchical material template

The process begins with template fabrication of solid polymer fractal structures up to the third order of hierarchy. These two-dimensional fractal patterns are sequentially transmitted to the reflective LCoS chip, which is illuminated with UV light from a light emitting diode array. Each image is projected through a reduction lens onto the surface of the photosensitive resin (Fig S2). The exposed liquid cures, forming a layer in the shape of the two-dimensional image, and the substrate on which it rests is lowered, reflowing a thin film of liquid over the cured layer. The image projection is then repeated with the next image slice forming the subsequent layer.

To further expand the scalability of hierarchical feature sizes. An addressable spatial light modulator (SLM) in coordination with an optical scanning system to produce large-scale parts with ~ 10 μm resolution. The SLM is imaged onto a photosensitive material with an optical system that has the ability to scan the image over a large area. The pattern change on the SLM is coordinated with the scanning rate of the optical system. The pattern is updated as the image is moved via the scanning optics to effectively create a continuous image in the photosensitive material. This scanned image is much larger than a single image of the SLM. This enables small feature sizes on a large area. This technique allows for fabrication of the polymer template hierarchical material with hundreds of millimeters in size and multi-scale features down to the $\sim 10\mu\text{m}$ scale, which is uniquely suitable for fabricating hierarchical materials.

S1.2 Nickel alloy lattice fabrication

All chemicals for electroless nickel deposition were obtained from Sigma-Aldrich (Milwaukee, WI) unless otherwise indicated. To prepare each sample for electroless nickel deposition, the sample was first cleaned by immersing in an acetone bath with gentle agitation for 2 minutes. After rinsing in deionized (DI) water for 30 seconds, the sample was immersed in a 1 M NaOH solution for 2 minutes followed by rinsing in DI water for 1 minute. Next, the sample surface was sensitized by immersing in an aqueous solution containing 3.5 g/L of $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and 5.0 mL/L 37% HCl for 2 minutes followed by rinsing in DI water for 30 seconds. After sensitization, the sample surface was activated with palladium catalyst by immersing in an aqueous solution containing 0.5 g/L PdCl and 4.0 mL/L 37% HCl for 2 minutes followed by rinsing in DI water for 30 seconds. After activation, the surface was metallized with nickel by immersing in a bath consisting of equal parts by volume of EdgeMet Electroless Nickel Kit A and B (Buehler, Lake Bluff, IL). This step was performed at 80°C with vigorous stirring. The coating thickness was controlled by varying the time of the metallization step between 1–20 minutes. Following this step, the sample was removed and rinsed with DI water for 30 seconds and allowed to air dry. Subsequently, the polymer interior was removed via NaOH base solution decomposition at 80°C for 24-48 hours. This last step yielded hollow-tube nickel-phosphorus hierarchical lattice with thickness ranging from 80nm-1µm.

S 1.3 Nickel phosphorous characterization

X-ray diffraction (XRD). XRD was carried out using Bruker AXS D8 ADVANCE X-ray diffractometer with X-ray source from Cu $K\alpha$ radiation operated at 40 kV and 40 mA. Several layers of film-like Ni-P samples prepared by electroless plating were restacked and sandwiched

in between two layers of Kapton tapes. 2 θ scan was conducted from 10° to 90° with 0.02° steps and 2 s counting time per step.

.3 Mechanical Testing

Nanoindenter and Instron based on design parameters. With face centered cubic orientation, the flat punch tip approaches the flat surface of the material. Monotonic and cyclical uniaxial compression experiments were performed on nanolattices via G200XP Nanoindenter (Agilent Technologies). In the first set of experiments, structures were compressed uniaxial to ~50% strain at a rate of $\mu\text{m/s}$ to determine their yield stress and overall deformation under load characteristics. Unloading rather than loading moduli were used to mitigate the possible effects of loading imperfections such as misalignment and partial initial contact.

To study the tensile stress and strain of the hierarchical lattices, we designed a custom dog bone sample for tensile testing. The dog bone gauge length consists of an architected lattice built in between two rectangular beams which serve as grips for conducting the tensile test. Typical dog bone grips match the lattice cross sectional area to ensure an even distribution of force to the lattice. Each grip measures 3mm in length to have a sufficient amount of material to grip for repeatable testing. Our Instron 5943 mechanical tester is set up with a 100N load cell () and custom made side action grippers. Tests are conducted at an extension rate of 10 $\mu\text{m/sec}$ and monitored with a video camera (Dino-Lite). During the test, we record the force exerted by the Instron along with the resulting lattice deformation.

S1.4 Finite Element Simulation

The finite element simulations were performed with the NIKE3D fully implicit solver, developed at Lawrence Livermore National Laboratory [1]. A tetrahedral mesh was generated from the same STL files from which the micro-stereolithography masks were derived. The average element size was roughly $0.15d$, where d is the diameter of the first layer struts, which produced 2-4 million element meshes. Using a linear elastic materials model based on the properties of photopolymerized HDDA, the architectures were compressed between two thin plates with moduli several orders of magnitudes stiffer than the polymer. A linear displacement of up to 30% strain was applied to the upper plate, while the lower one remained fixed. Each simulation ran until significant buckling was observed in the lattices.

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S2 Analysis of the failure modes of hierarchical lattices

S2.1 First order Octet hierarchical cell

As described in the main article, the hierarchical stretch dominated lattice comprised of fractal unit cells where members of unit cells comprised of stretch dominated lattice. These unit cells in each level has a regular octahedron core with regular tetrahedral affixed to each face, which remains stretch dominated in each hierarchical orders separated by at least one order of magnitude in length-scale. The unit cell's cubic symmetry (detailed in Figure S1) results in a material with nearly isotropic properties. In fractal or self-similar configurations, the failure of fractal lattice originate from a combination of four failure mechanisms with each level of hierarchy: Local buckling of wall, Euler Buckling of first order tube, first order struts yielding/fracture, Euler Buckling of second order lattice.

On the macroscale, under uniaxial compressive loading, the relative compressive stiffness and yield strength of these structures theoretically show linear scaling relationships: $E/E_s \sim (\rho/\rho_s)$ and $\sigma/\sigma_s \sim (\rho/\rho_s)$. Consider a n^{th} order lattice of density ρ_n and effective Young's modulus E_n^* , with the cell walls made of a lattice material of density and effective Young's modulus, ρ_{n-1} and E_{n-1}^* , respectively. The Young's modulus E_n can be expressed as $E_n^*/E_{n-1}^* = k_n(\rho_n/\rho_{n-1})^{m_n}$. The effective property of the lattice material is described at each level of hierarchy, which serves as an equivalent plastic model that described the mechanical property of unit cells.

We define the aspect ratios of hierarchical strut members, $d(i)/l(i)$, where $d(i)$ is the diameter of struts, $l(i)$ is the length of the strut, i is the hierarchical rank of the structural elements within the material.

For a hollow tube hierarchical stretch dominated lattice, the relative density (the volume percentage of the metallic material contained in the lattice volume envelope in Fig S1 is given by

The relative density of a hierarchical lattice with total number of hierarchy n is given by:

$$\rho = \prod_{i=1}^n \rho_{i-(i-1)}$$

Where i is the hierarchical rank, ρ_i is the relative density of material made up of the i^{th} hierarchical level. For the second order strut $\rho_{21} = 6\sqrt{2}(\frac{d_2}{l_2})^2$, $\rho_{10} = 6\sqrt{2}\pi(\frac{d_1}{l_1})(\frac{t}{l_1})$

$$\rho_{20} = 36\sqrt{2}\pi(\frac{d_2}{l_2})^2(\frac{d_1}{l_1})(\frac{t}{l_1})$$

1) Critical stress at the first order of hierarchy

$$\sigma_f > \sigma_b$$

For the base material considered in this study, the fracture stress of electroplated Nickel-phosphorous constituent material is 1.78-2.0GPa, Young's Modulus 97-105 GPa. The critical buckling stress at the first order of hierarchy is given by

$$\sigma_{1-0 (shell)} = \frac{2E_0 t}{\sqrt{3(1-v^2)}}$$

$$\sigma_{1-0 (euler)} = \frac{\pi^2 E_0 I_1}{k^2 l_1^2 A_1}$$

$$\sigma_{1-0 (yielding)} = \sigma_f$$

Where A and I are the cross sectional area of the first order hollow tube and the area moment of inertia $A = \pi d_1 t$, $I = \frac{1}{8} \pi d_1^3 t$, $k=0.5$ for rigid joints.

To calculate the equivalent stress at the second order of hierarchy as a result of the first order critical stress, the second order lattice, as illustrated in Figure S 2, is modeled as a [110] lattice with forces applied at [110] direction. The equivalent stress σ_{2-1} corresponding to three failure modes at the first order strut is given by:

$$\sigma_{2-1}(\sigma_{1-0 (shell)}) = \frac{2\pi E_0 \sin\theta_1 (\frac{t}{l_1})^2}{\sqrt{3(1-v^2)} \cos^2\theta_1}$$

$$\sigma_{2-1}(\sigma_{1-0 (euler)}) = \frac{E_0 \pi^3 (\frac{d_1}{l_1})^3 (\frac{t}{l_1}) \sin\theta_1}{2 \cos^2\theta_1}$$

$$\sigma_{2-1}(\sigma_{1-0 (f)}) = \frac{\pi \sin\theta_1 (\frac{d_1}{l_1}) (\frac{t}{l_1})}{\cos^2\theta_1} \sigma_f$$

Global effective strength: $\sigma[\sigma_{1-0 (shell)}] = \frac{4\sqrt{2}\pi E (\frac{t}{l_1})^2 (\frac{d_2}{l_2})^2 \sin\theta_1 \sin\theta_2}{\sqrt{3(1-v^2)} \cos^2\theta_1 \cos^2\theta_2}$

Global effective strength: $\sigma[\sigma_{1-0 (euler)}] = \sqrt{2}\pi^3 E (\frac{d}{l})^3 (\frac{t}{l}) (\frac{d_2}{l_2})^2 \frac{\sin\theta_1 \sin\theta_2}{\cos^2\theta_1 \cos^2\theta_2}$

where θ_2 is the neutral axis of the second order struts with respect to the X axis of the global coordinate.

2) Critical stress at the second order of hierarchy

Second order hierarchical struts buckling occurs when the critical buckling stress $\sigma_{2-1 (Euler)} < \sigma_{2-1 (\sigma_{1-0(f)})}$ that is, when the minimal global stress is lower than minimal global stress when the first order struts fracture. We define the intermediate stress at the second order of hierarchy as:

$$\sigma_{2-1 (euler)} = \frac{\pi^2 E_{2-1} I(2)}{k^2 l_2^2 A_2}$$

$$I_2 = \frac{\sqrt{2}}{12} d_2^4$$

$$A_2 = \frac{\sqrt{2}}{2} d_2^2$$

Therefore, the critical buckling stress at the struts at the second order of hierarchy is given by:

$$\sigma_{2-1 (euler)} = \frac{\sqrt{2} \pi^2}{9} \left(\frac{d_1}{l_1}\right) \left(\frac{t}{l}\right) \left(\frac{d_2}{l_2}\right)^2$$

The global stress induced by second order struts buckling can be derived from the relation between the global stress and the critical Euler buckling stress at the second order related as:

Global Effective strength:
$$\sigma[\sigma_{2-1 (Euler)}] = \frac{2}{9} \pi^2 \left(\frac{d_1}{l_1}\right) \left(\frac{t}{l_1}\right) \left(\frac{d_2}{l_2}\right)^4 \frac{\sin \theta_2}{\cos^2 \theta_2}$$

3) Failure modes

The failure mode of the hierarchical lattice is the one associated with the lowest collapse strength for a given set of structural configurations at the zero, first and second order of hierarchy. We can find the critical transitional values for activation of failure modes at each successive hierarchical level.

Transition between first order yielding/fracture and first order Euler buckling. The failure modes determination criteria can be expressed by the following simple relations

$$\sigma_{1-0 (shell)} = \sigma_f$$

$$\sigma_{1-0 (Euler)} = \sigma_f$$

$$\min\{\sigma[\sigma_{1-0 (euler)}], \sigma[\sigma_{1-0 (shell)}], \sigma_f\} = \sigma[\sigma_{2-1 (Euler)}]$$

Therefore, the overall operational failure mode of hierarchical lattices can be traced as a combinations of multi-scale parameters $d_1/l_1, t/d_1, d_2/l_2$ set out by the above critical loading conditions. For given values of electroless plated Nickel phosphorous properties, these parameters can be determined as:

First order shell buckling:

$$\frac{t_c}{d_{1c}} = \frac{\sqrt{3(1-v^2)}\sigma_s}{2E}$$

And

$$\frac{d_2}{l_2} > \sqrt{\frac{18\sqrt{2}}{\sqrt{3(1-v^2)}}} \left(\frac{t}{d_1}\right)$$

First order Euler Buckling

$$\frac{d_{1c}}{l_{1c}} = \sqrt{\frac{2\sigma_s}{\pi^2 E}}$$

Second order Euler Buckling

$$\frac{d_2}{l_{2c}} < \sqrt{\frac{18\sigma_s}{\pi^2 E}}$$

And

$$\frac{d_2}{l_2} < \sqrt{\frac{18\sqrt{2}}{\sqrt{3(1-v^2)}\pi}} \left(\frac{t}{d_1}\right)$$

Given these bounds, the predicted failure modes for each of the hierarchical lattices can be described in Fig. X and Table .

Based on the analysis above, for as-fabricated solid polymer hierarchical lattices, there are three failure modes: fracture of the first order strut, Euler Buckling of the first order struts, shell

buckling of the first order struts and Euler buckling of the second order struts, which are plotted in the map below. The critical wall thickness to diameter ratio of the first order hollow tube

Failure mode space and optimal scenario

Optimal scenario: In order to optimize the mechanical efficiency of a hierarchical structure, the material should be on the edge of both instabilities. The global yield stress is determined by the lowest yield stress as a competition between different failure modes.

$$\sigma_m = \min [\sigma_{b2}, \sigma_{b1}, \sigma_{s1}]$$

Equating (10) and (14), and combine with (1) we plot the maximum strength corresponding to each t/d_1 and $d_2/12$ pair in Figure 3E

Table S1. Summary of fractal lattice with a variety of hierarchical combinations

Fig. S1 Fabrication of third order fractal lattice

Fig. S2 Design of fractal lattice library

Fig. S3 Volumetric FEM and continuum plasticity model for hierarchical lattice properties

Fig. S4 The results of FEM investigation showing relative compressive stiffness as a function of relative density of octet-truss lattices in different loading orientations.

Fig S5 Optimized fractal hierarchy

Fig. S6 Loading and unloading curves of hierarchical architected materials

Fig. S7 Loading and unloading of metallic sponges

Fig. S8. Loading and unloading of ceramic composite materials

Movie S1. Uniaxial compression of 5 % solid polymer (HDDA) bend-dominated lattice (played at 4x speed).

Movie S2. Uniaxial compression of a 9.8 % solid polymer (HDDA) stretch-dominated lattice (played at 4x speed).

Movie S3. Uniaxial cyclic compression of a stretch-dominated ultralight hollow-tube Ni-P lattice ($\rho=11.8 \text{ kg/m}^3$)

Framework Builder - Configurable Fractal Lattice Generator

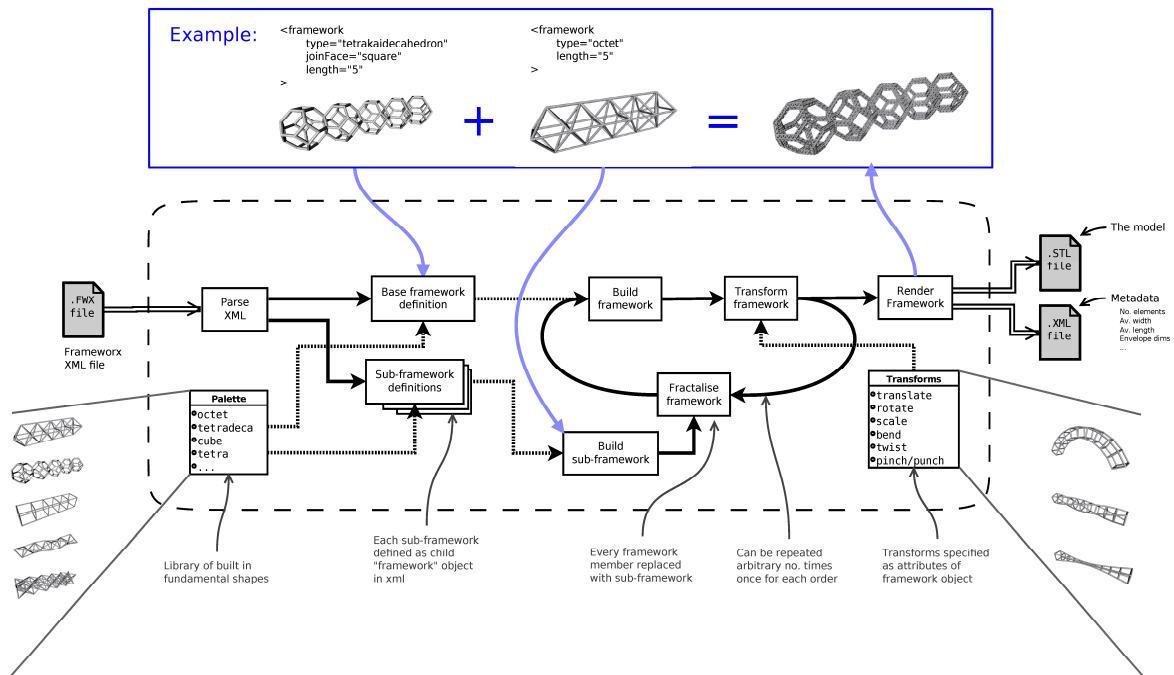


Figure S3

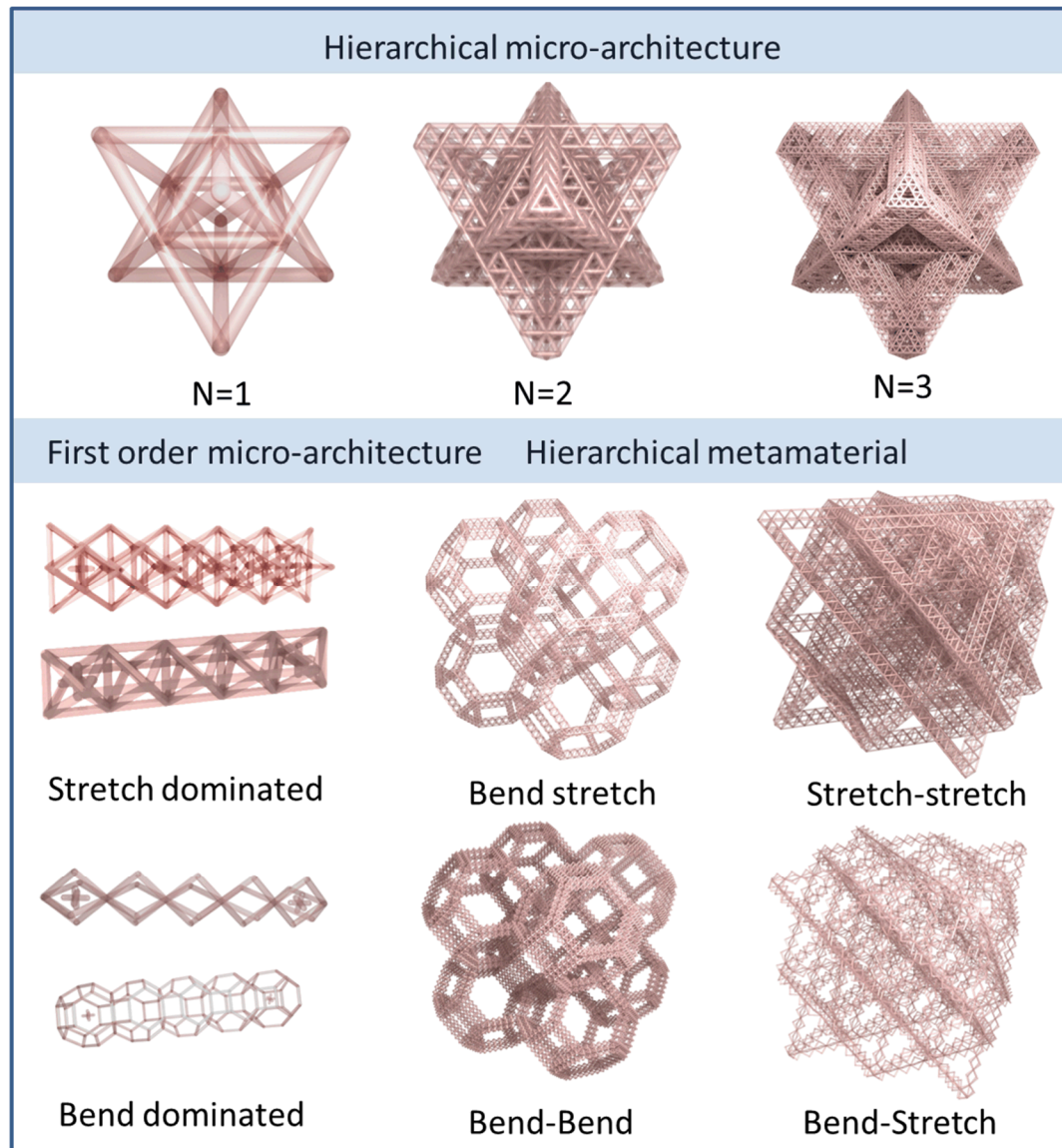


Figure S4

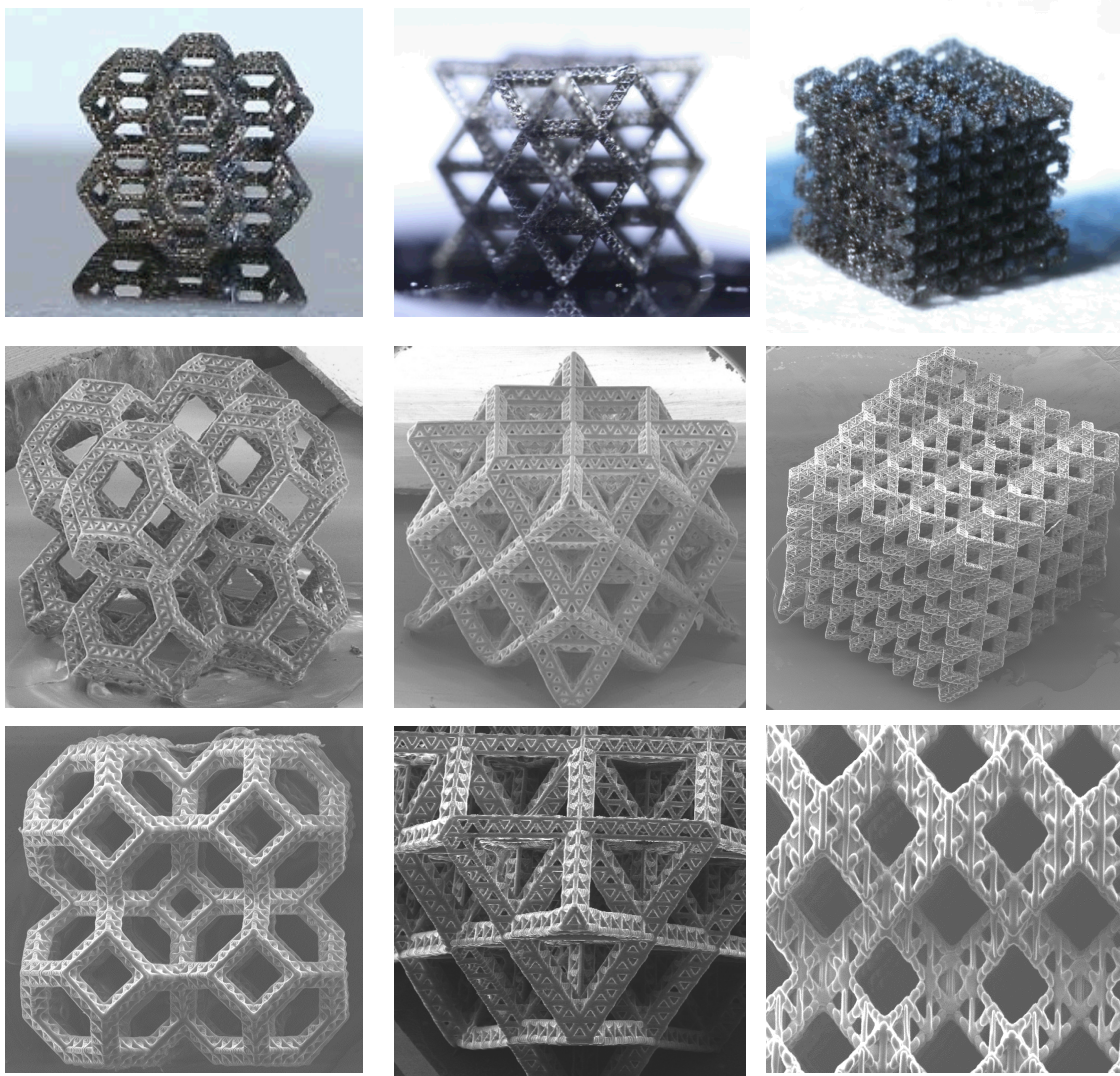
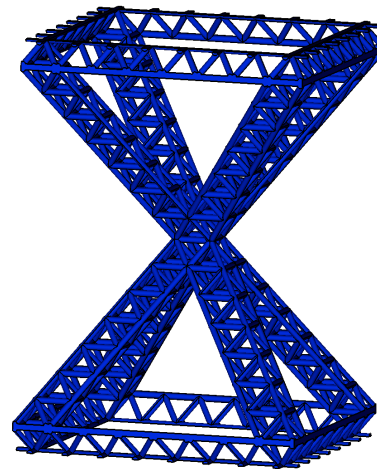
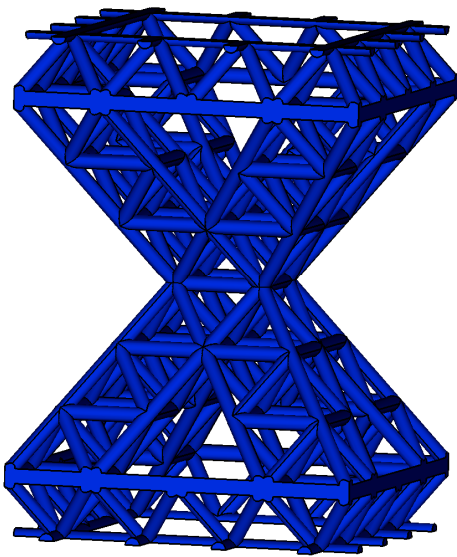
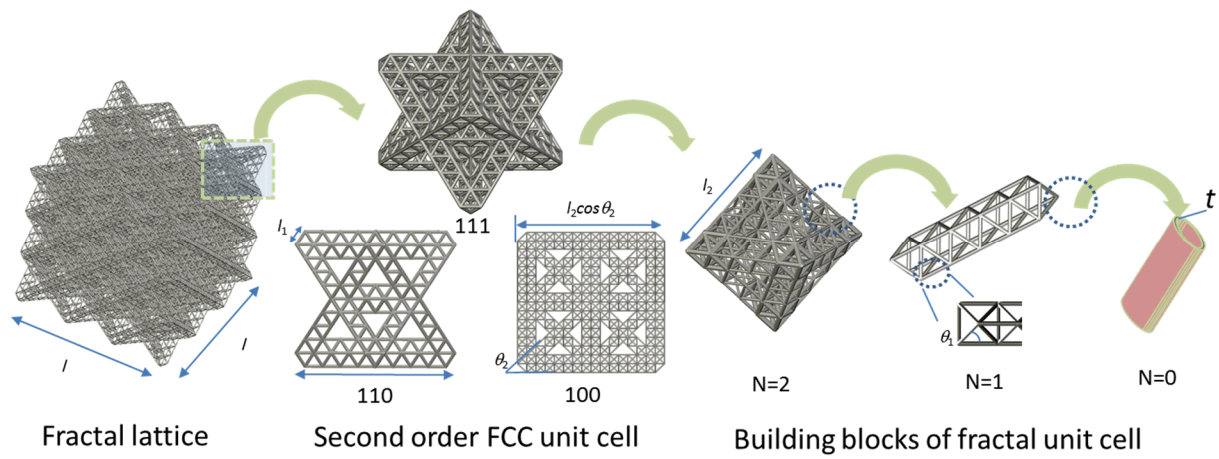
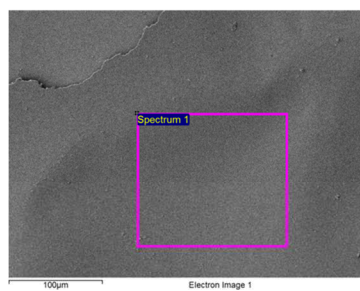


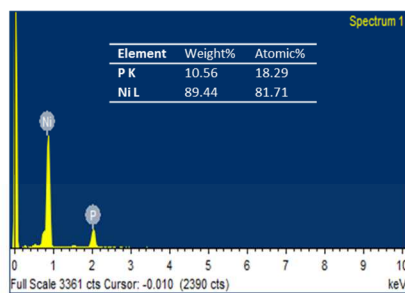
TABLE S1

Lattice Material and Architecture	Yield strength (KPa)	Relative Density	Wall thickness t (nm)	t/d_1	d_2/l_2	Failure Mode
Second Order Octet Micro lattice	110	1.58E-3	200	0.0134	0.15	First order tube buckling (optimal)
	70	1.6E-3	200	0.01	0.18	First order tube buckling
	63	1.61E-3	200	0.008	0.2	First order tube buckling
	205.8	2.66E-3	200	0.017	0.18	First order tube buckling (optimal)
	120	2.5E-3	100	0.014	0.19	First order tube buckling (optimal)
	70	2.6E-3	100	0.024	0.15	Second order tube buckling
	75	2.5E-3	100	0.025	0.15	Second order tube buckling
	94	2.4E-3	100	0.012	0.2	First order tube buckling
	50.2	8.22E-4	70	0.03	0.16	Second order filament buckling
	17.2	5.6E-4	70	0.014	0.1	Second order filament buckling
	30.5	5.43E-4	70	0.009	0.12	First order tube buckling Optimal
	25.6	4.68E-4	70	0.008	0.12	First order tube buckling
	7.48	2E-4	70	0.006	0.09	First order tube buckling
	4.8	2.2E-4	70	0.006	0.1	First order tube buckling

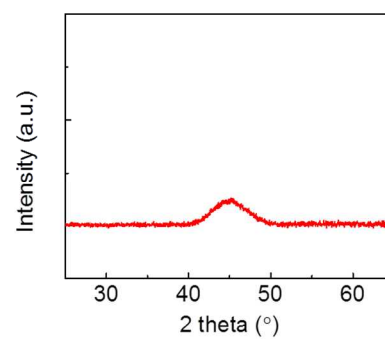




A



B



C